

Rain Zhang

Full-Stack Developer & Product Engineer · Simon Fraser University

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EXPERIENCE

Software and IT Systems Specialist

MNT Realty · Vancouver, BC, Canada

Aug 2026 – present

- Sole software engineer for the company: own every internal system from requirements and architecture to hosting, deployment and maintenance.
- Scoped and rebuilt the MNT Realty platform from the ground up: a new public website, an owner portal for residents, and a staff admin console in one application.
- Chose the workflows worth replacing and built MNT Control Center, bringing the office's scattered tools into one place, in Next.js, Node.js and PostgreSQL.
- Set up organization sign-in and role-based access with Microsoft Entra ID, the Graph API and OAuth 2.0.
- Automated the slowest recurring office work: sorting incoming email, looking up owner information, and answering staff questions from company documents.

Full-Stack Engineer Intern

FEITIAN Technologies · International Dept. · Beijing

Sep – Dec 2025

- Defined and delivered three production systems for the post-quantum FIDO2 program: a public developer platform, a Rust software security key, and a customer demo site.
- Wrote a CTAP2.1 authenticator in Rust that Linux presents as a virtual USB security key, letting browsers test ML-DSA credentials before hardware existed.
- Deployed with Docker on Linux servers and Google Cloud Run; GitHub Actions ran tests, builds and a daily FIDO metadata refresh.
- Worked with hardware and security engineers so the tools matched real devices; still maintaining all three after the internship ended.

PROJECTS

MNT Realty Platform

Next.js · TypeScript · Node.js · PostgreSQL · Vercel
· mntrealty.vercel.app

Aug 2026 – present

- One application serving three audiences: the public website, an owner portal for residents, and a staff admin console, each on its own subdomain.
- Host-based routing gives each surface its own chrome and session cookie; every read and write goes through one data-store contract.
- Navigation, page guards and file routes read one access table, so no role ever learns what another can see.

WebAuthn Developer Platform

Python · Flask · Docker · liboqs · webauthnlab.tech · repository

Sep 2025 – present

- Public tool for FIDO2 developers, built because the team had none: register and sign in with real or virtual authenticators, decode responses, and search the FIDO Alliance metadata service.
- Added ML-DSA-44, -65 and -87 to a fork of Yubico's python-fido2 through liboqs, without breaking the classical paths.
- About 120 server test files plus frontend and post-quantum suites run in CI; a daily GitHub Action re-verifies the metadata snapshot.

SKILLS

Languages

Python, TypeScript,
JavaScript, Rust, C, C++, Java

Web

React, Next.js, Node.js, Flask,
Tailwind CSS, HTML, CSS,
REST APIs

Data and cloud

PostgreSQL, Docker, Google
Cloud, Vercel, GitHub
Actions, Git, Linux

Security

WebAuthn, FIDO2, CTAP2.1,
OAuth 2.0, Microsoft Entra ID,
post-quantum cryptography
(ML-DSA, liboqs)

Testing

pytest, Vitest, Playwright, CI
on GitHub Actions

EDUCATION

Simon Fraser University

Bachelor of Science,
Computer Science
Vancouver, BC, Canada

Sep 2023 – Apr 2027 · CGPA
3.44 / 4.33

Dean's Honour Roll, Fall 2024
and Summer 2025